Noise and Outliers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 7: inspect types, units and source records.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Calculate mean and median. Try the recall before opening notes.

Your learning target

I can calculate **mean** and **median**.

I can **investigate** an **unusual value** before deleting it.

I can explain how a summary can hide **variation**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 17: **Noise** and **Outlier**s

Mistakes and **unusual value**s should be **investigate**d, not automatically erased.



THINKER MISSION

Decide whether four odd records are errors or real cases.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What is the total of 2, 3 and 4?

This week's end goal

By the end, I can calculate **mean** and **median** and **investigate** an **unusual value** before changing it.

STUDENT LESSON · 2 OF 18

Notice the evidence

Investigate the supplied Noise and Outliers case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **noise** and **outlier**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects.

Keep all values while investigating whether 18 is a genuine long activity or a recording error.

STUDENT LESSON · 3 OF 18

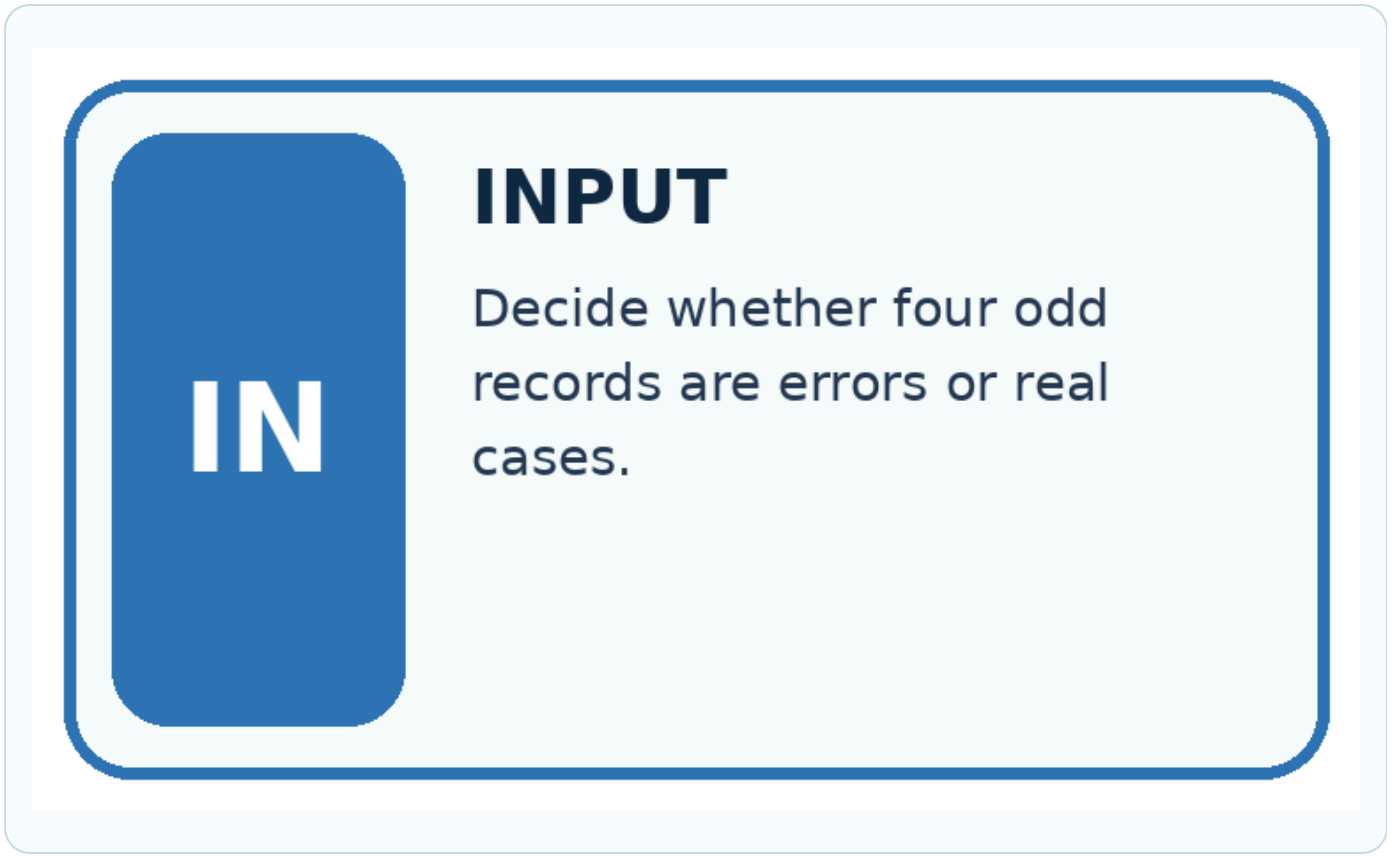
See how it works

> Follow the mechanism for Noise and Outliers. Connect each labelled input, deciding step and result.

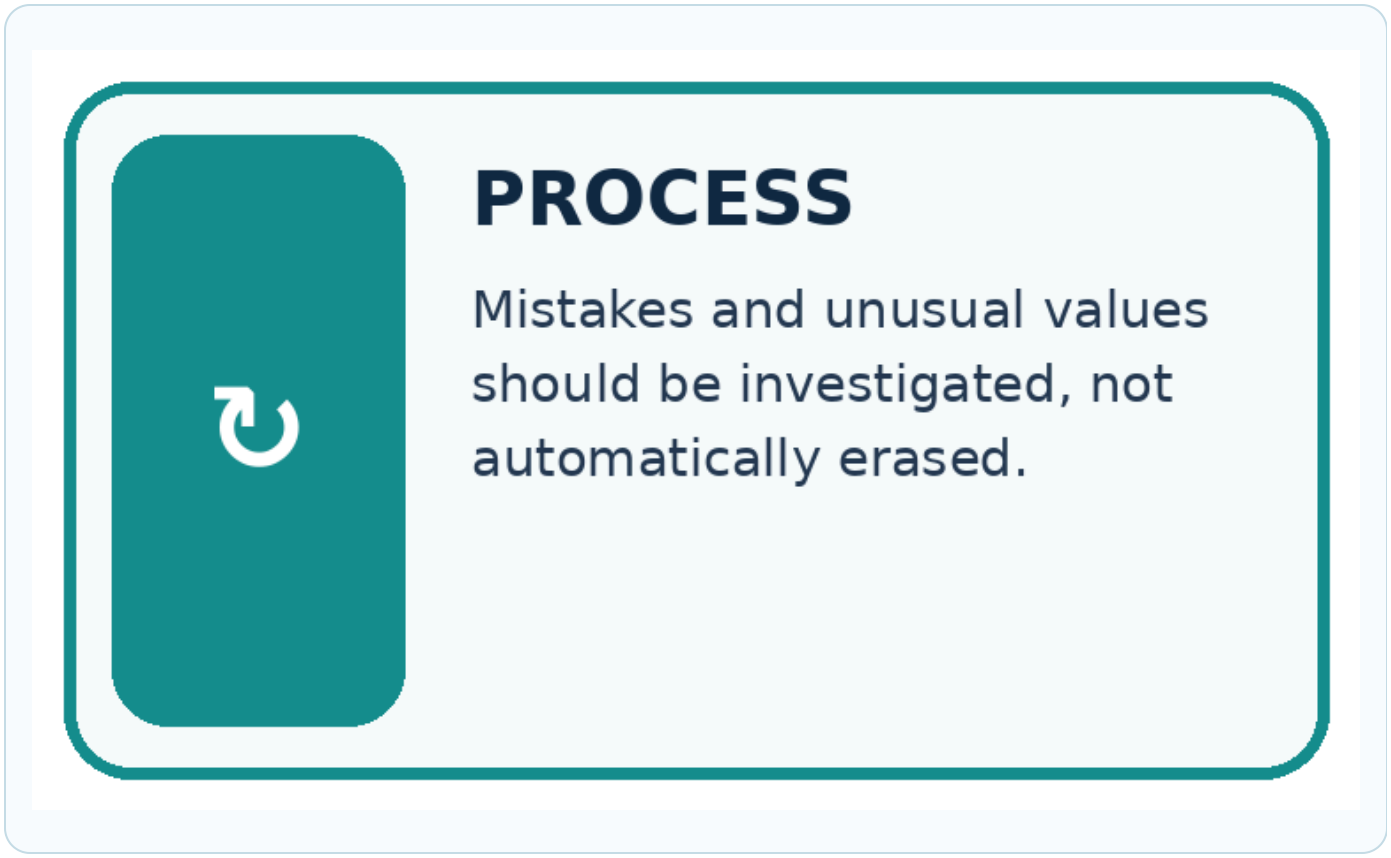
VISUAL MECHANISM

Input → Process → Output

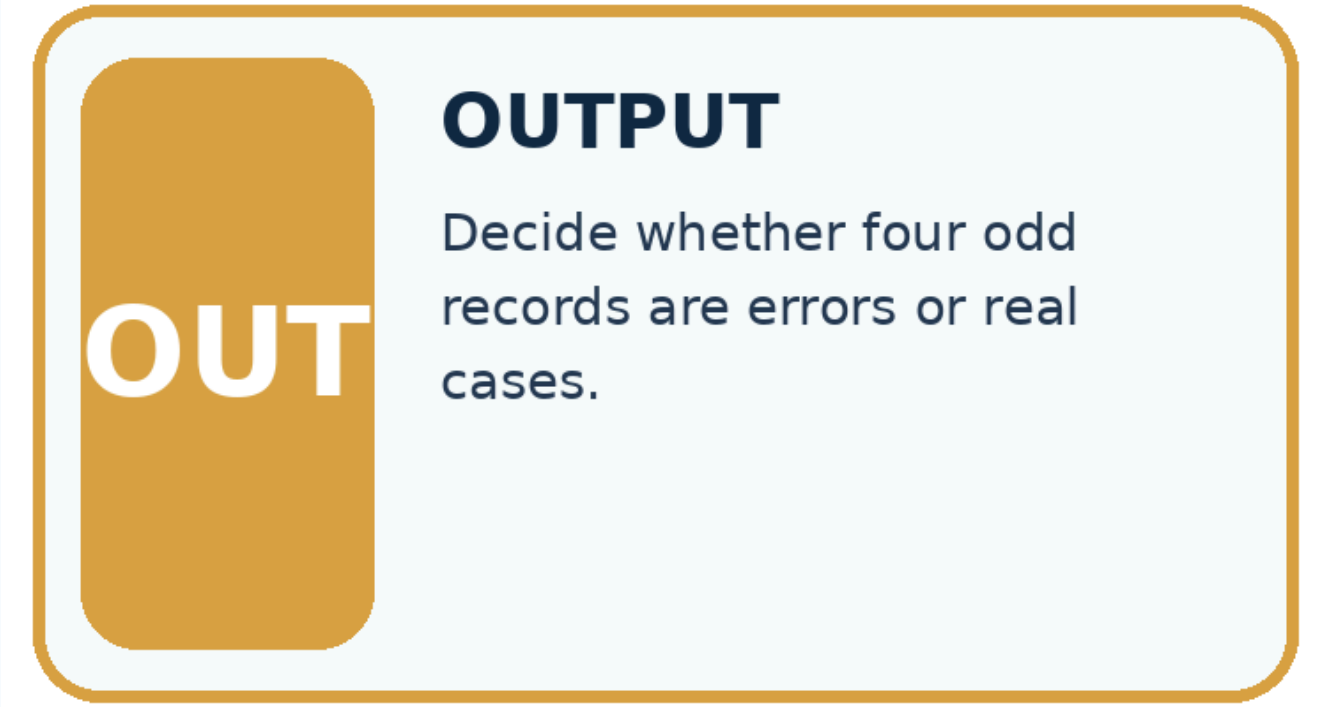
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Decide whether four odd records are errors or real cases.

↓


PROCESS
Mistakes and unusual values should be investigated, not automatically erased.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

```
values = [2, 3, 3, 4, 18]
ordered = sorted(values)
mean = sum(values) / len(values)
median = ordered[len(ordered) // 2]
print(mean)
print(median)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Sorted data	2, 3, 3, 4, 18	Middle is 3
Mean	30 divided by 5	6.0
Typical middle	Third of five values	3

Read each row from left to right and connect the evidence to the stated rule.

The script prints 6.0 and 3. sorted returns a sorted list without changing the original. This **median** expression works for a nonempty odd-length list only. For four sorted values, average the second and third.

Do not use this odd-length shortcut for every dataset.

Use two summaries for two questions

The **mean** shares the total equally: 2 + 3 + 3 + 4 + 18 is 30, and 30 divided by five is 6. The **median** asks for the middle value after sorting: the third value is 3.

The long duration 18 pulls the **mean** upwards but leaves the middle value at 3. A difference between the summaries is a clue to inspect the data, not permission to remove the inconvenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

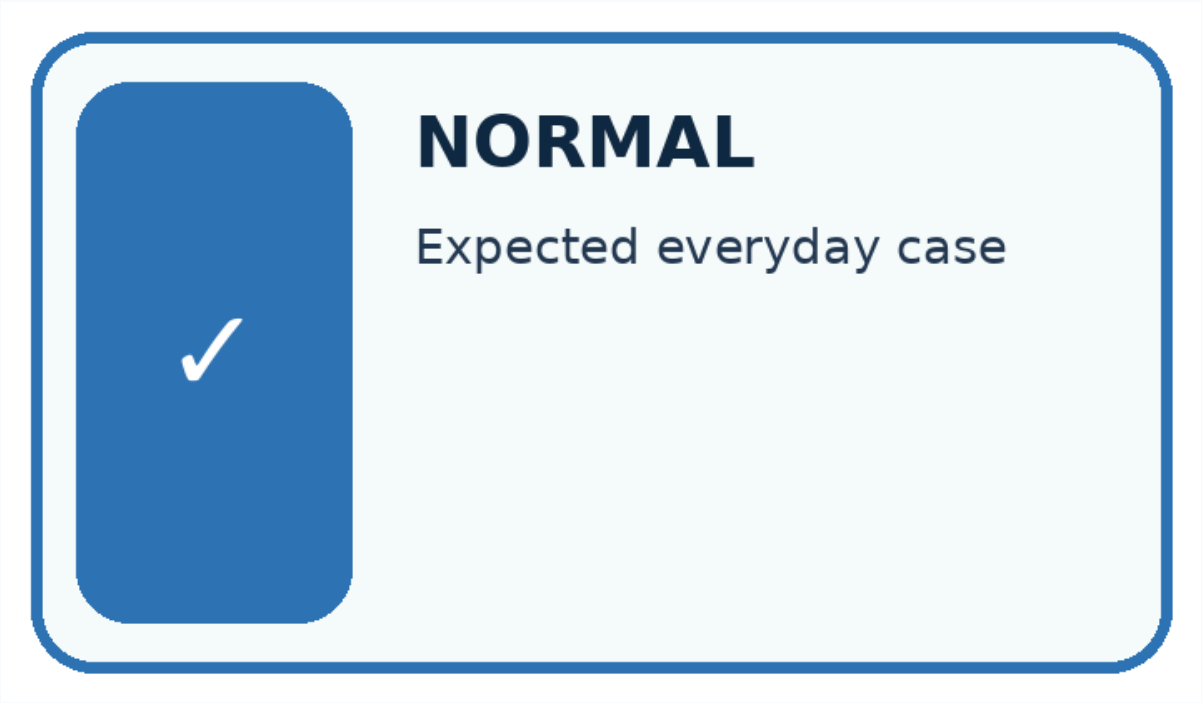
> Compare the supplied cases for Noise and Outliers. Identify the detail that changes the conclusion.

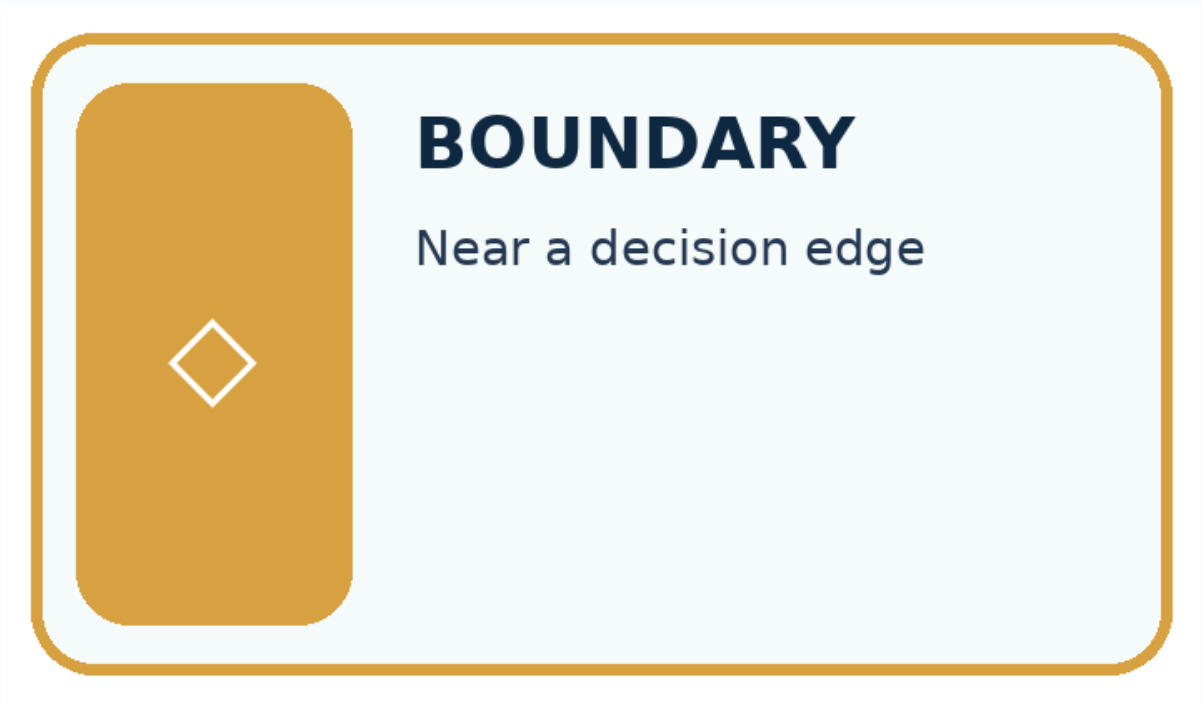
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

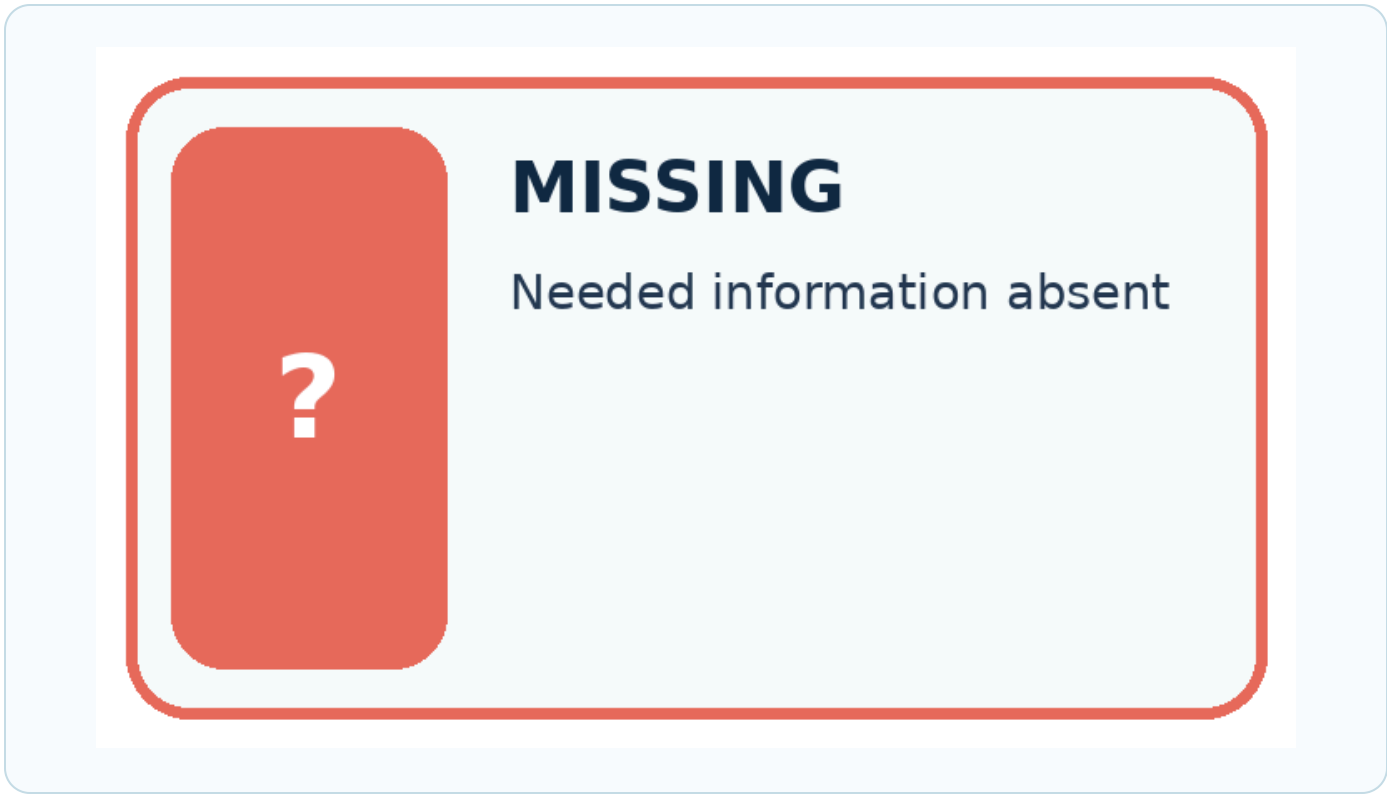
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: An authorised log confirms 18 was a transcription error for 3.

Predicted result

Prediction: Corrected **mean** is 3 minutes and **median** is 3.

Why the result follows

Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

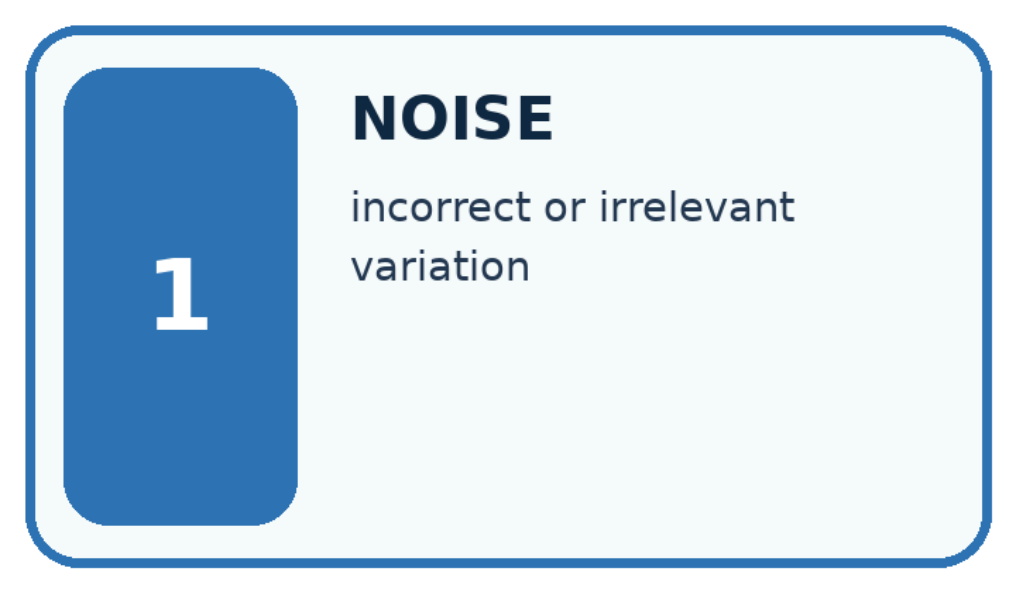
Words that unlock the idea

> Use the vocabulary of Noise and Outliers to explain a concrete decision, rather than only memorising definitions.

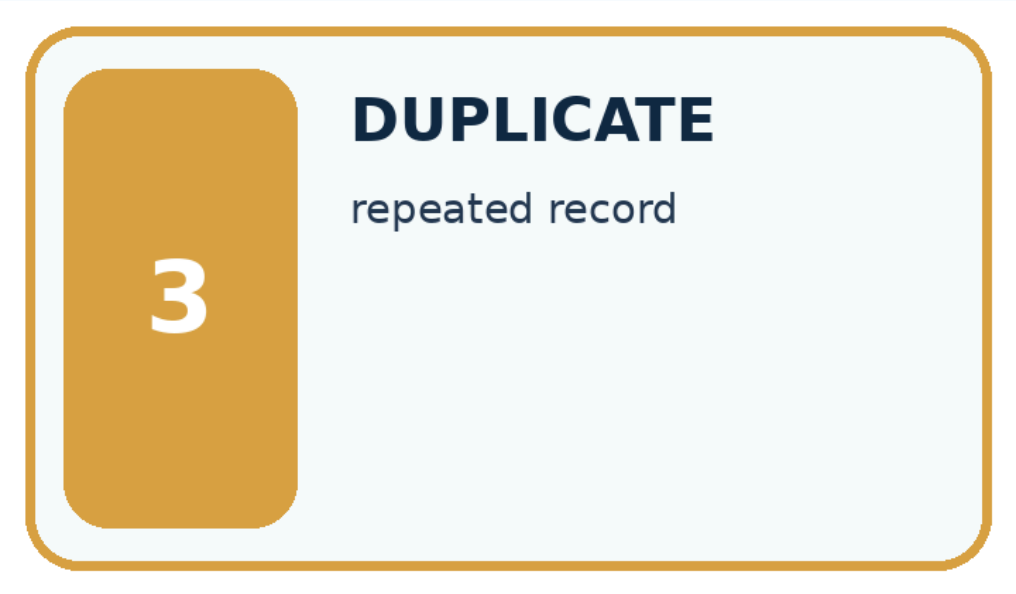
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
OUTLIER
value far from most others


3
DUPLICATE
repeated record



Word	My own example	Not this / boundary
noise	Variation or interference that obscures the information relevant to a task.	
outlier	An observation far from a typical pattern under a stated comparison; it is not automatically an error.	
duplicate	A repeated record or copy; copies are not independent new observations.	
investigate	Collect and examine evidence to answer a question rather than assuming a cause.	

Words used in the closer explanation

Word	Meaning
Mean	Sum of numeric values divided by their count
Median	The middle of sorted values or mean of the two middle values
Unusual value	A value far from most others in a dataset
Variation	Differences among recorded values

STUDENT LESSON · 6 OF 18

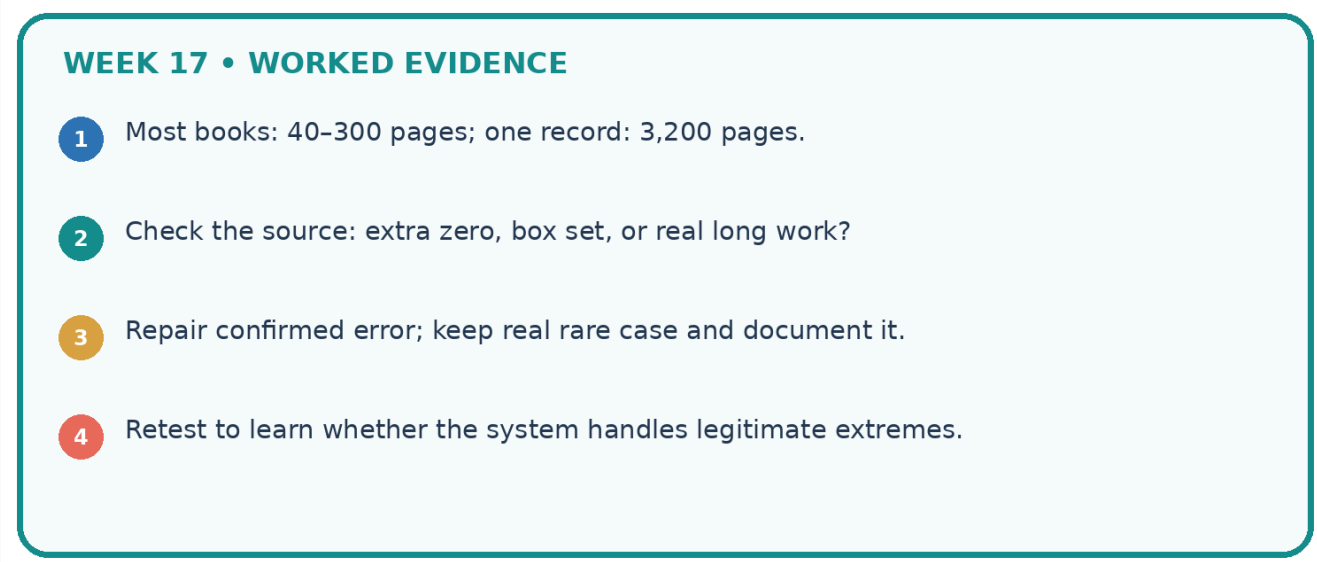
Follow a worked example

> Follow the worked trace for Noise and Outliers, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Most books: 40–300 pages; one record: 3,200 pages.

2. Check the source: extra zero, box set, or real long work?

3. Repair confirmed error; keep real rare case and document it.

4. Retest to learn whether the system handles legitimate extremes.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Replace 18 with 8. Calculate the new mean and median.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Noise and Outliers

A fictional library records five book-**process**ing times in minutes.

The facts we will use

Supplied facts

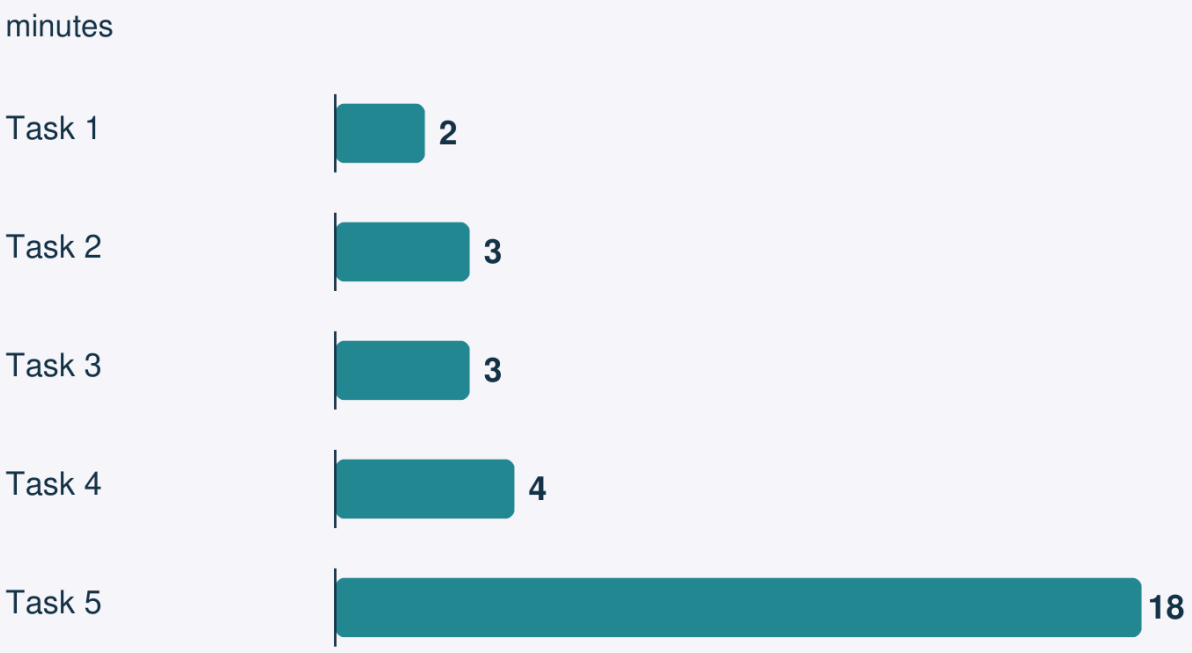
Sorted times: 2,3,3,4,18.

Supplied facts

All are recorded measurements; no **evidence** yet shows 18 is a mistake.

AI ACADEMY / AI-2/17

Retain unusual recorded measurements



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/17

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mean

Sum 30; divide by 5: mean 6 minutes.



2. Median

The middle of five sorted values is 3 minutes.



3. Investigate 18

Check the source record and whether an unusually slow task occurred.



4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Mean	Sum 30; divide by 5: mean 6 minutes.
2. Median	The middle of five sorted values is 3 minutes.
3. Investigate 18	Check the source record and whether an unusually slow task occurred.
4. Document repair	Change a value only if evidence supports a correction; preserve the raw record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

What this example does not establish

Keep the stated limit

Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

Keep the stated limit

Read the labelled picture: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Keep the stated limit

Change one thing: An authorised log confirms 18 was a transcription error for 3. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Noise and Outliers claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Every unusual value is bad data that should be deleted. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

> **A dataset is empty. Is reporting a mean of 0 justified?**

STUDENT LESSON · 8 OF 18

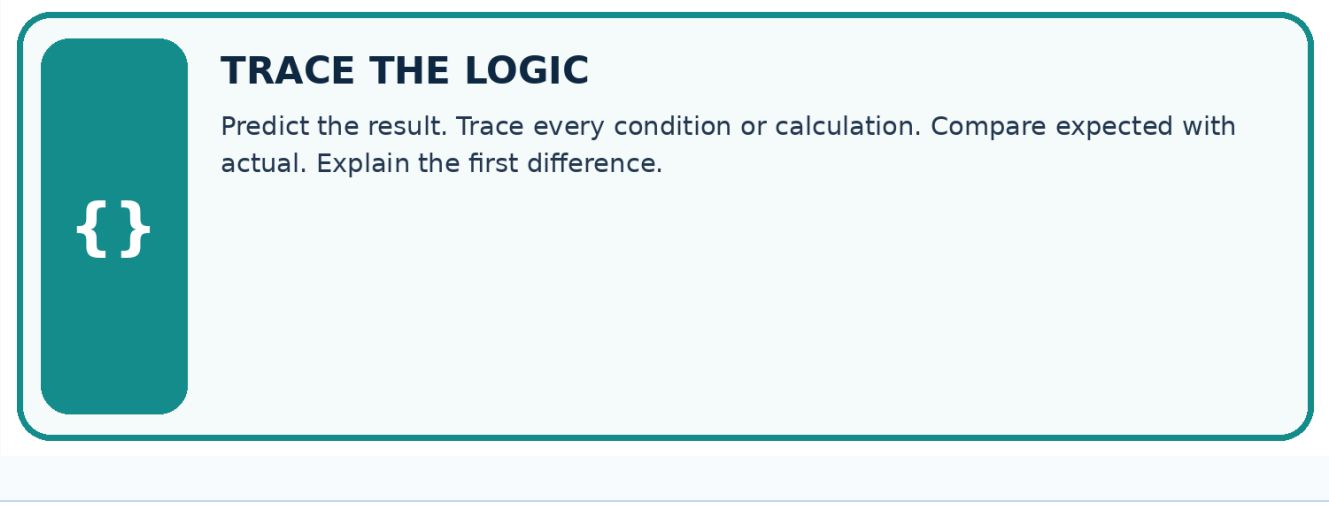
Practise together

> Practise Noise and Outliers with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Taking the middle item of an unsorted list can give the wrong **median**. For [18, 2, 4, 3, 3], the middle item is 4, but the sorted **median** is 3. Sort first. For an empty list, there is no **mean** or **median** under this lesson's rules.

Practise with less help: Compute two medians

Use workbook W2 for the connected example "**Mean median** and **unusual value**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

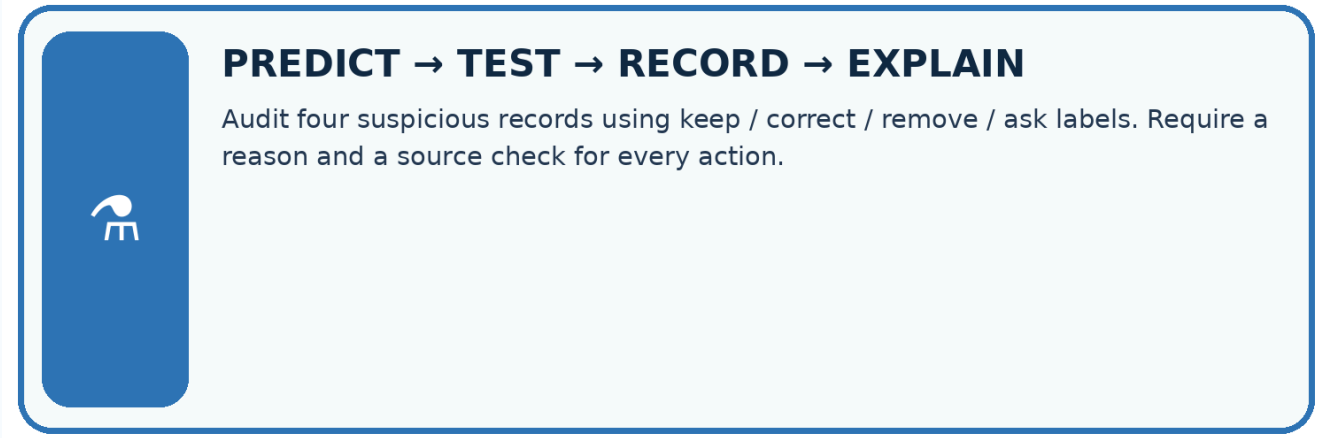
Run the investigation

> Investigate Noise and Outliers. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit four suspicious records using keep / correct / remove / ask labels. Require a reason and a source check for every action.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

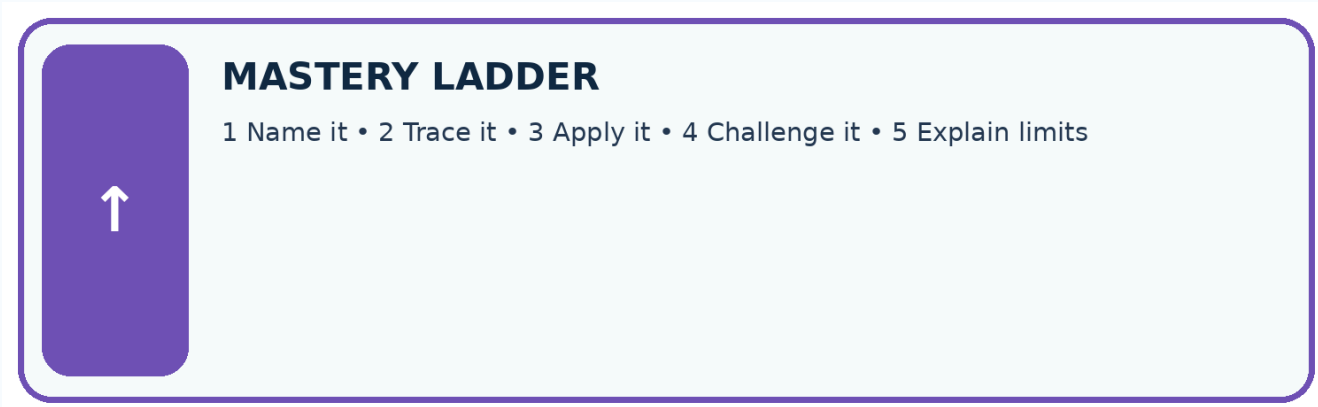
Build your understanding

Move from recall to tracing, diagnosis and transfer in Noise and Outliers. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

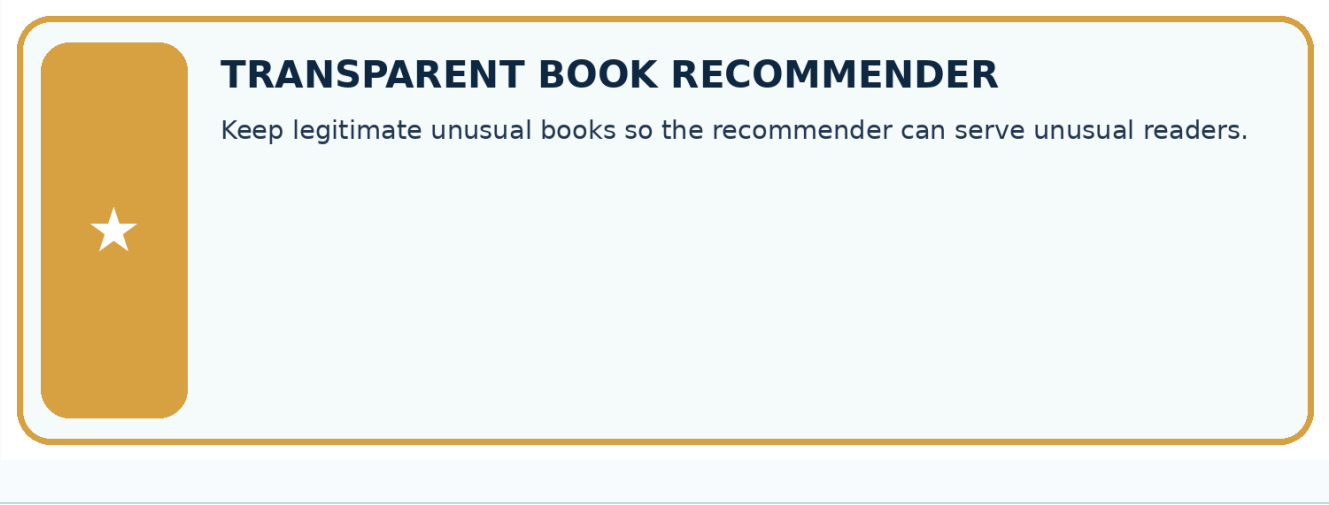
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Keep legitimate unusual books so the recommender can serve unusual readers.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

Show what you know

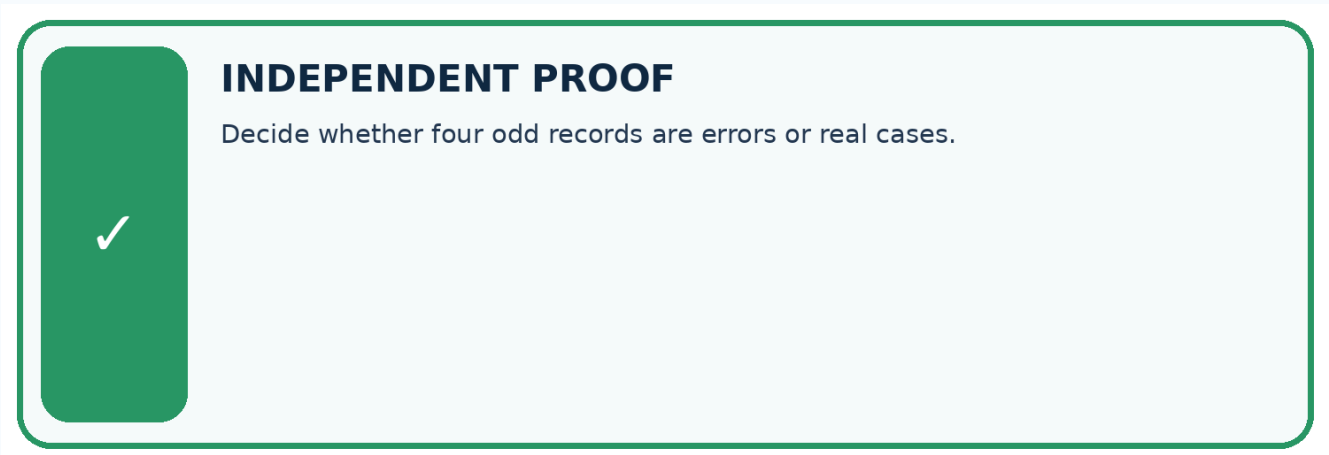
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Decide whether four odd records are errors or real cases.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Noise and Outliers case for supported practice. Repeated examples do not become fresh assessments.

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Recorded lengths are 90,95,100,105,900 pages.

The question

> Is 900 automatically wrong?

Worked explanation

No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

An outlier needs investigation

Evidence or stage	Value or result
Book A	90
Book B	95

Evidence or stage	Value or result
Book C	100
Book D	105
Book E	900

Recorded lengths are 90,95,100,105,900 pages.

> Is 900 automatically wrong?

No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compare summaries independently

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to **investigate**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Keep legitimate unusual books so the recommender can serve unusual readers.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Noise and Outliers, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional library records five book-**process**ing times in minutes.

By the end, I can	How I will show it
Calculate mean and median	Explain the deciding step, show a trace or test, and state one limit.
Investigate an unusual value before deleting it	Explain the deciding step, show a trace or test, and state one limit.
Explain how a summary can hide variation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → **investigate** → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Noise and Outliers without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate **mean** and **median**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional library records five book-**process**ing times in minutes.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Change a value only if **evidence** supports a correction; preserve the raw record.

1. **Median**

2. **Investigate** 18

3. Document repair

4. **Mean**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Document repair Change a value only if **evidence** supports a correction; preserve the raw record. The **mean** uses every value and is pulled upward by 18.

The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: An authorised log confirms 18 was a transcription error for 3. Which conclusion is supported by the supplied facts?

1. Corrected **mean** is 3 minutes and **median** is 3.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Corrected **mean** is 3 minutes and **median** is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Repair a missing building block

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

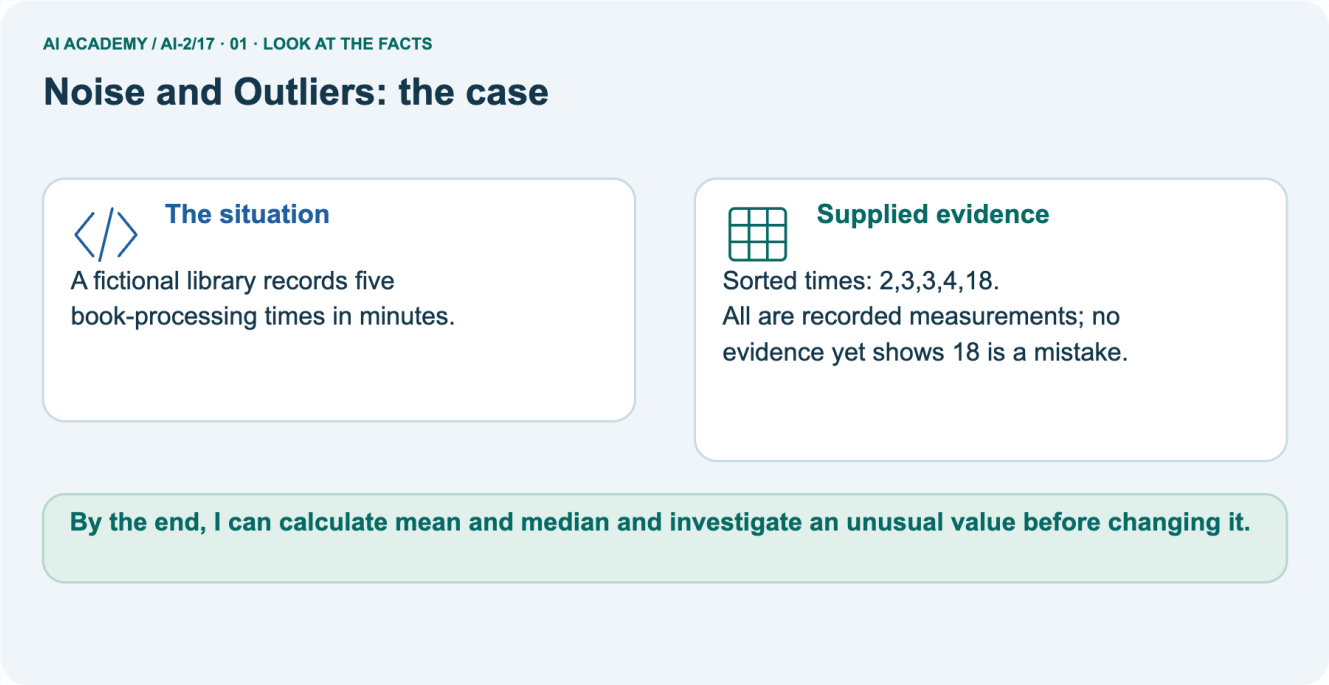
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Noise and Outliers graphic using its labels, deciding evidence and relevant limitation.

01. Noise and Outliers: the case

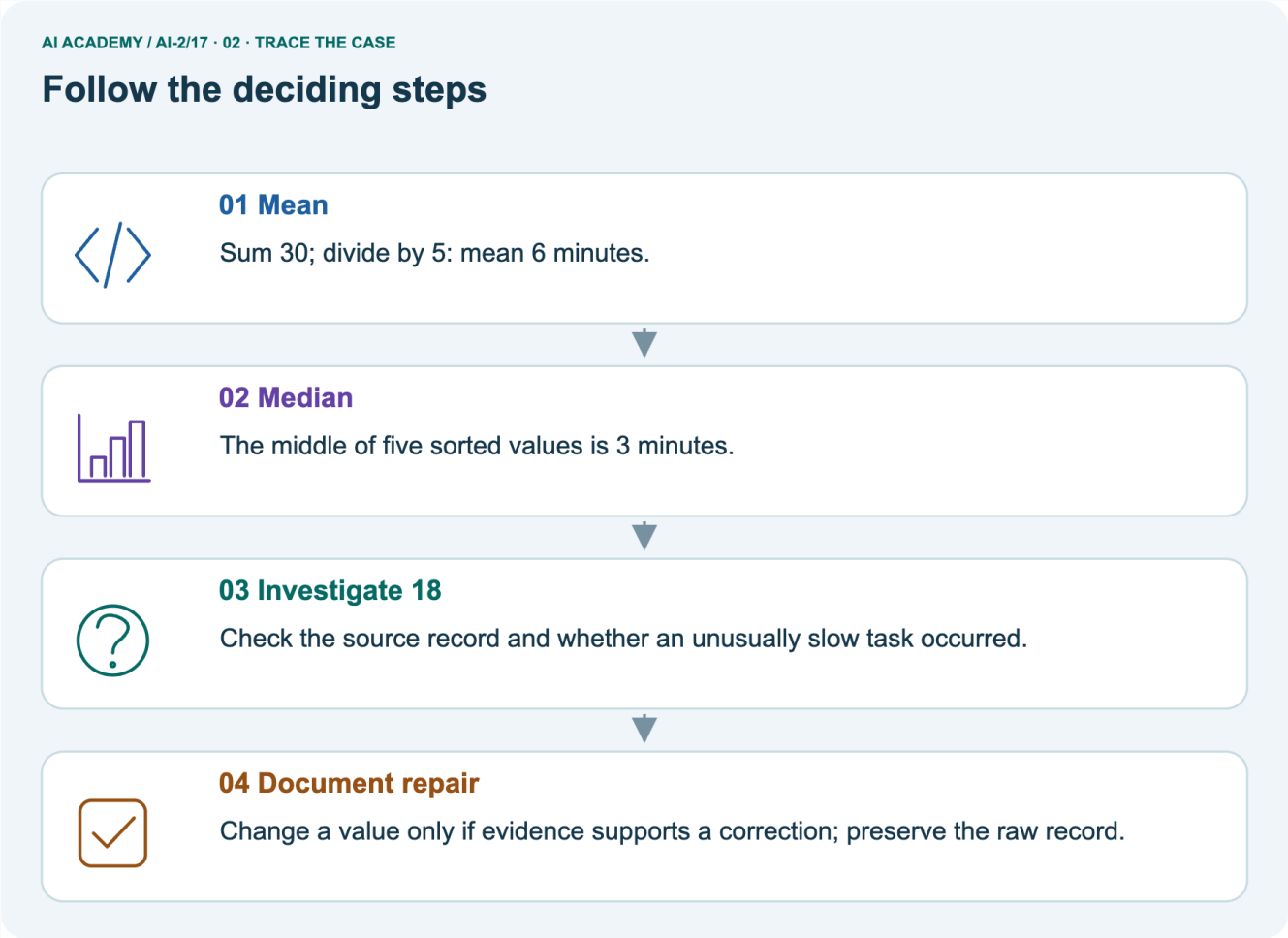


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional library records five book-**process**ing times in minutes. Supplied facts: Sorted times: 2,3,3,4,18. All are recorded measurements; no **evidence** yet shows 18 is a mistake.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

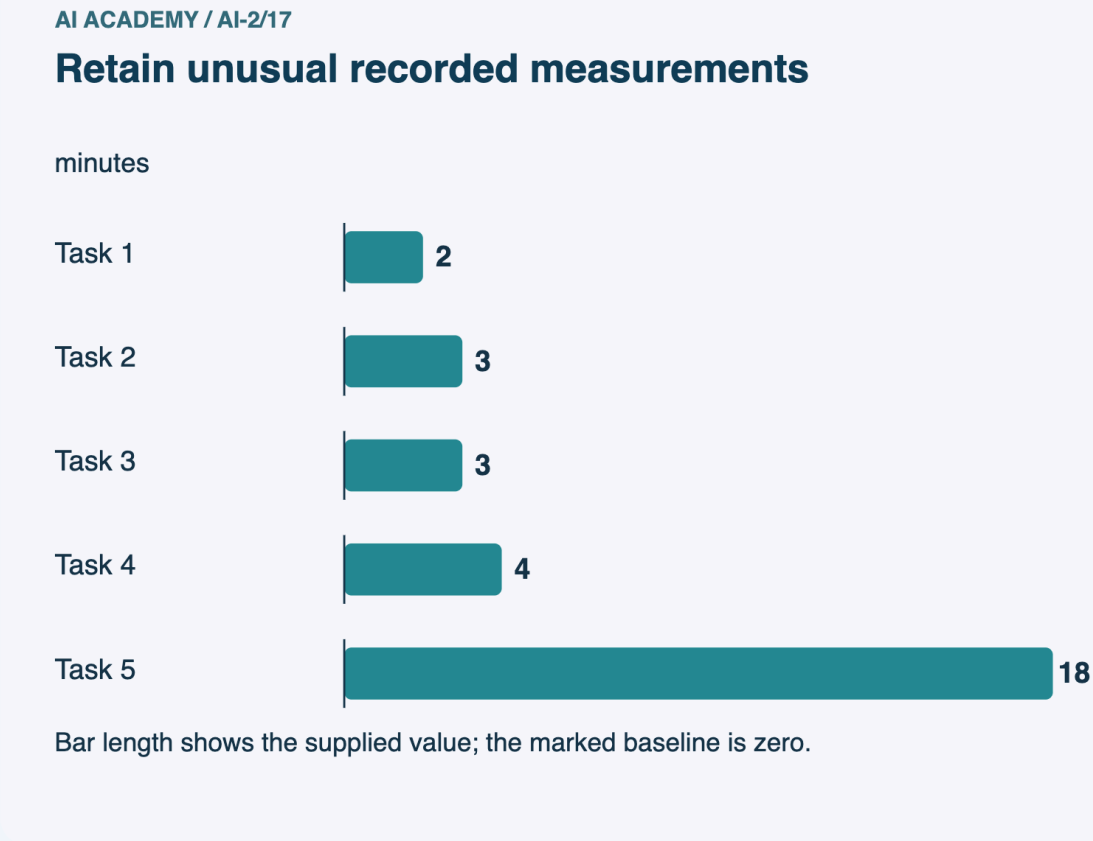


Follow the labelled picture with the matching explanation. Open the vector diagram

Mean: Sum 30; divide by 5: **mean** 6 minutes. → **Median**: The middle of five sorted values is 3 minutes. → **Investigate** 18: Check the source record and whether an unusually slow task occurred. → Document repair: Change a value only if **evidence** supports a correction; preserve the raw record.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Retain unusual recorded measurements

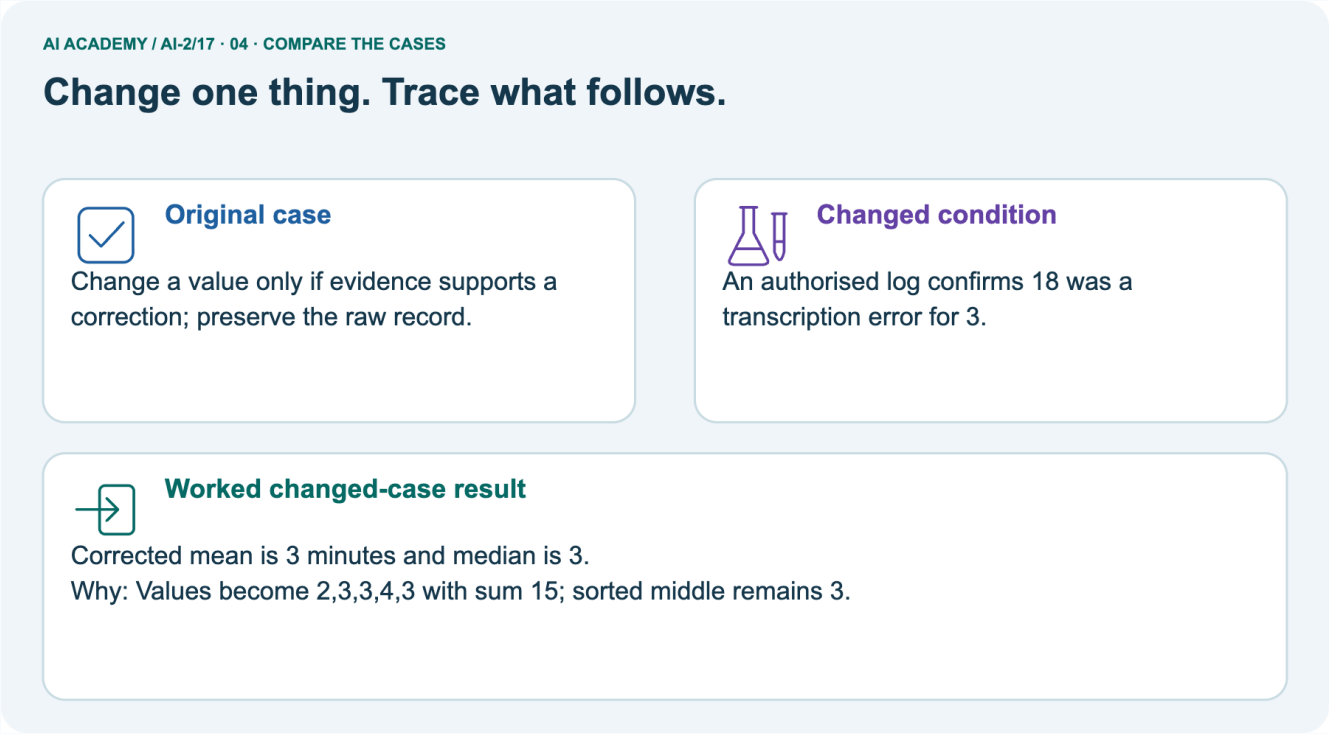


Follow the labelled picture with the matching explanation. Open the vector diagram

Retain unusual recorded measurements. Units: minutes. Task 1: 2. Task 2: 3. Task 3: 3. Task 4: 4. Task 5: 18.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



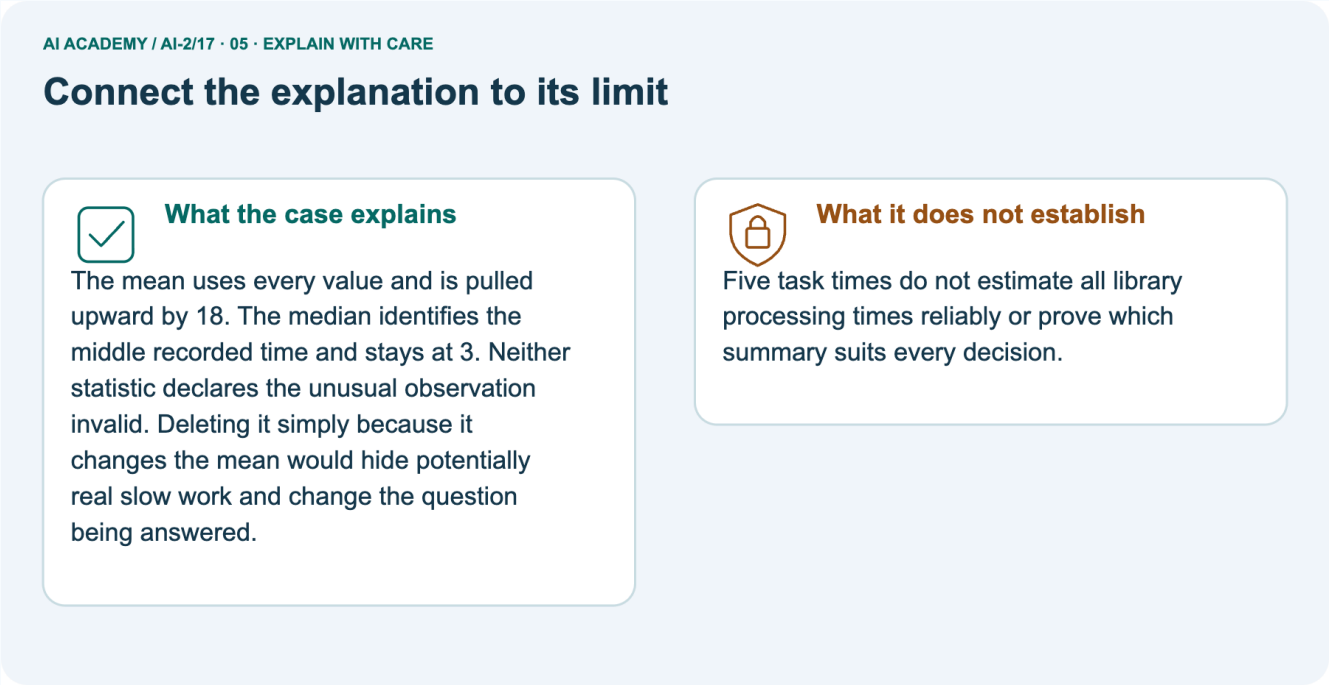
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Change a value only if **evidence** supports a correction; preserve the raw record.
Changed condition: An authorised log confirms 18 was a transcription error for 3.

Expected result: Corrected **mean** is 3 minutes and **median** is 3. Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

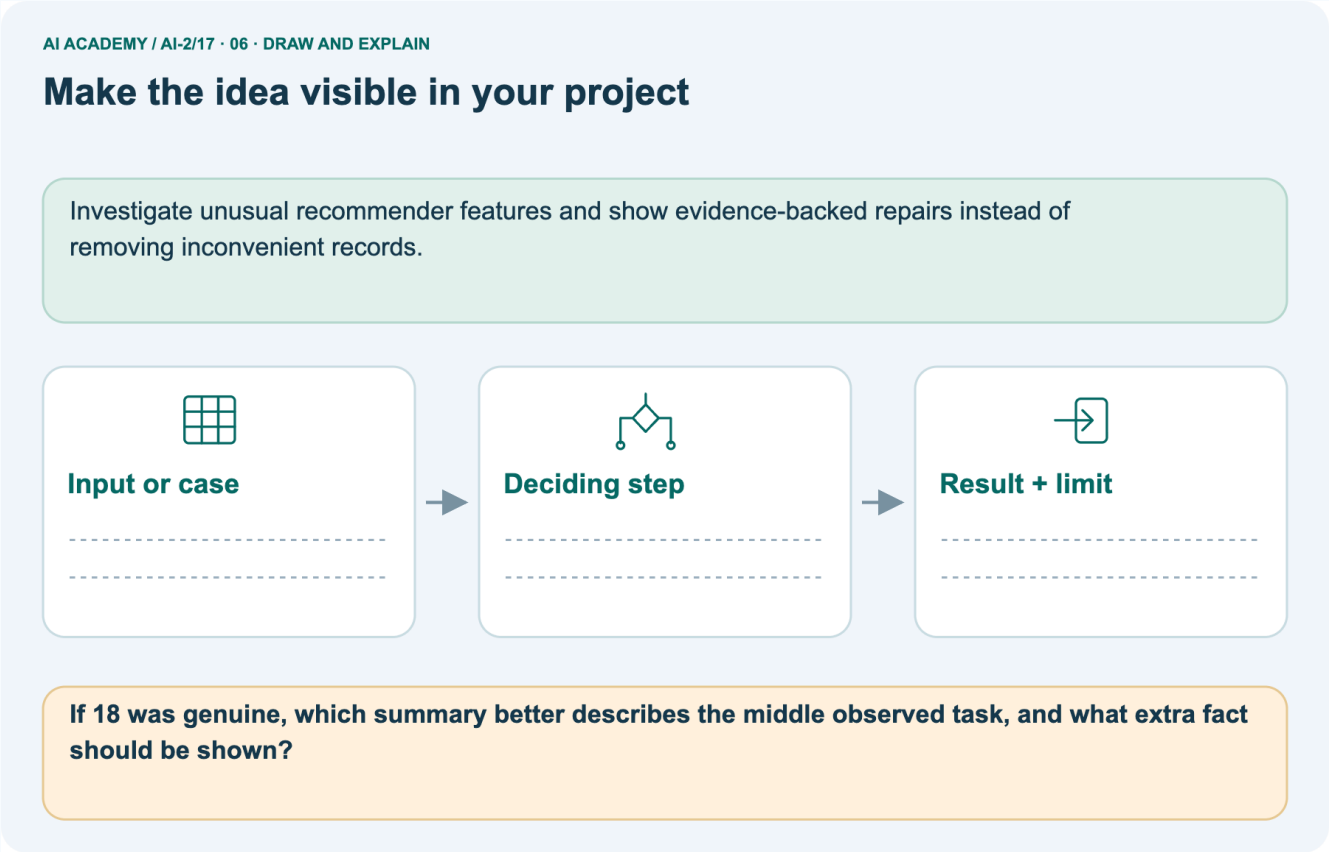
The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Limit: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records. Draw your **input**, deciding step, result and limit.

Supported question: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.